Assessing Energy Injection in Decoupled Friction Compensation for Differential Haptic Interfaces

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***Abstract*—Differential transmissions for haptic interfaces trade reduced reflected inertia for coupled friction across joints due to internal load-sharing. Conventional friction compensation treats each joint independently through a diagonal model, an assumption evaluated primarily through single-axis experiments. Through a case study on a custom differential haptic interface, we show that the decoupled formulation produces significant residual energy errors under multi-axis motion. By applying the Work–Energy Theorem over closed periodic trajectories, we isolate non-conservative frictional errors from conservative dynamics and zero-mean unmodeled effects. Specifically, a decoupled Stribeck friction compensator either underestimates dissipation, degrading transparency, or injects energy into the system, indicating active controller behavior and a potential passivity violation. The residual torque exhibits severe directional anisotropy, with errors exceeding 8.6 Nm in one velocity quadrant but remaining below 4.2 Nm in the opposite quadrant despite comparable speeds. This asymmetry alters the transmission's dissipation profile, highlighting the need for coupled compensation strategies that account for transmission mechanics.**

***Index Terms*—haptics and haptic interfaces, dynamics, calibration and identification, mechanism design, friction compensation, passivity**

I. INTRODUCTION

Haptic interfaces commonly employ differential transmissions to concentrate actuator mass at the base, reducing reflected inertia and improving backdrivability [1], [2]. Their mechanical coupling, however, redistributes internal loads and alters the equivalent friction experienced by one joint, which may vary with the loading condition of another, even when its own velocity remains unchanged. This behavior is particularly relevant in haptic systems, where uncompensated friction degrades transparency [3], and excessive compensation may induce active controller behavior, potentially compromising passivity and stability [4], [5], [6].

Classical analytical models [7], [8] and recent adaptive approaches [9], [10], [11], [12] describe friction as a function of the local joint state. Data-driven methods have begun capturing coupled behaviors in multi-degree-of-freedom (DoF) systems [13], [14], [15], yet joint-wise formulations remain the dominant paradigm. Conventional identification procedures reflect this decoupled approach, typically tuning models through isolated single-axis experiments, which suits independent joints but provides little evidence that the assumption holds under load redistribution. The energetic consequences of load-dependent friction coupling and their impact on transparency and passivity have not been experimentally characterized in differential haptic interfaces. In this work, we focus on *SENTIR* (Fig. 1), a custom differential haptic interface, as a case study.

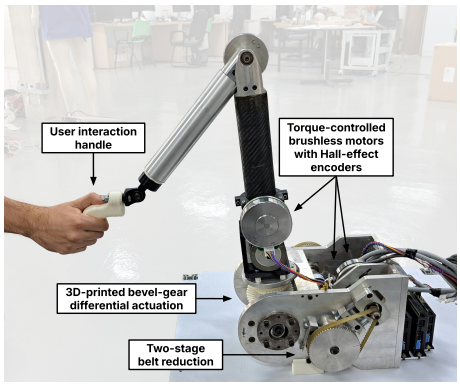


Fig. 1. SENTIR differential haptic interface. The distal joint was removed to isolate the two-DoF differential stage studied in this work.

We assess the decoupled friction assumption using an energy-based methodology that separates non-conservative compensation effects from conservative rigid-body dynamics. Based on the Work–Energy Theorem, we derive a residual energy metric that quantifies the net mechanical energy exchanged over periodic trajectories. Since conservative forces integrate to zero over a closed cycle, this metric isolates friction modeling errors and enables comparisons across trajectories by removing the influence of inertia and stiffness.

We experimentally evaluate our approach using five excitation trajectories (0.5 Hz) that progressively increase cross-axis excitation within the transmission. This specific low-frequency regime was selected to minimize velocity estimation phase lag and isolate continuous sliding friction from high-frequency inertial transients. Our results show that friction models

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identified through conventional single-axis procedures remain accurate during pure operating modes but exhibit significant errors during simultaneous multi-axis motion. Depending on the loading condition, the same decoupled compensator either underestimates dissipation, degrading transparency, or injects net energy, indicating a potential passivity violation.

The main contributions of this work are: **(i)** an energy-balance formulation that isolates friction compensation errors from conservative dynamics; **(ii)** experimental evidence that decoupled friction compensation can inject energy under coupled loading; and **(iii)** a characterization of directional anisotropy in residual torque as a function of velocity quadrant.

The remainder of the paper is organized as follows. We first establish the kinematic and dynamic properties of the differential transmission in Section II, forming the basis for the proposed energy-based analysis derived in Section III. After detailing the experimental platform and protocol in Section IV, we present the resulting energy injection and directional anisotropy in Section V. We conclude in Section VI with a discussion of the implications for passivity and an outline of future work on human-in-the-loop experiments.

II. KINEMATICS AND DYNAMICS

The defining characteristic of a differential transmission is the mapping between the motor space $\mathcal{M} \subseteq \mathbb{R}^2$ and the joint space $\mathcal{J} \subseteq \mathbb{R}^2$. We model the system as a two-DoF differential transmission, where the joint coordinates $\mathbf{q} = [q_1, q_2]^\top \in \mathcal{J}$ are related to the motor coordinates $\mathbf{q}_\mathrm{m} = [q_{\mathrm{m}1}, q_{\mathrm{m}2}]^\top \in \mathcal{M}$ through the constant transmission matrix $\mathbf{\Omega} \in \mathbb{R}^{2\times 2}$:

$$\mathbf{\Omega} = \frac{1}{2R}\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}, \tag{1}$$

where R is the transmission ratio, such that $\mathbf{q} = \mathbf{\Omega}\mathbf{q}_\mathrm{m}$. Enforcing power invariance between the actuator and joint spaces, $\boldsymbol{\tau}_\mathrm{m}^\top \dot{\mathbf{q}}_\mathrm{m} = \boldsymbol{\tau}^\top \dot{\mathbf{q}}$, yields the torque transformation

$$\boldsymbol{\tau}_\mathrm{m} = \mathbf{\Omega}^\top \boldsymbol{\tau} \implies \boldsymbol{\tau} = \mathbf{\Omega}^{-\top}\boldsymbol{\tau}_\mathrm{m} = R\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}\boldsymbol{\tau}_\mathrm{m}. \tag{2}$$

Substituting these mapped torques into the system's equations of motion, the motor-space dynamics are given by

$$\boldsymbol{\tau}_\mathrm{m} = \mathbf{\Omega}^\top \Big(\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \boldsymbol{\tau}_\mathrm{fric}\Big), \tag{3}$$

where $\mathbf{M}(\mathbf{q})$ is the inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ accounts for Coriolis and centrifugal effects, $\mathbf{g}(\mathbf{q})$ is the gravity vector, and $\boldsymbol{\tau}_\mathrm{fric}$ is the equivalent friction torque.

Equation (3) represents the ideal rigid-body dynamics. In physical differential transmissions, however, the torque equilibrium is perturbed by unmodeled transient effects, including state estimation delays and the interaction between structural compliance and mechanical backlash during velocity reversals.

Conventional decoupled formulations assume $\boldsymbol{\tau}_\mathrm{fric}$ to be an independent function of $\dot{\mathbf{q}}$, thereby ignoring the load redistribution imposed by $\mathbf{\Omega}$. Because direct time-domain evaluation of load-coupled friction is highly sensitive to these unmodeled high-frequency dynamics and observer phase lags, we introduce a cyclic energy-based evaluation that isolates and quantifies the modeling gap of the decoupled assumption.

III. CYCLIC ENERGY BALANCE

Consider the control law

$$\boldsymbol{\tau}_\mathrm{cmd} = \boldsymbol{\tau}_\mathrm{dyn} + \boldsymbol{\tau}_\mathrm{fb} + \boldsymbol{\tau}_\mathrm{comp}, \tag{4}$$

where $\boldsymbol{\tau}_\mathrm{dyn}$ compensates for rigid-body dynamics, $\boldsymbol{\tau}_\mathrm{fb}$ is the feedback term, and $\boldsymbol{\tau}_\mathrm{comp}$ is the friction compensation term. The residual torque is then

$$\boldsymbol{\tau}_\mathrm{net} = \boldsymbol{\tau}_\mathrm{cmd} - \boldsymbol{\tau}_\mathrm{comp}. \tag{5}$$

Assuming perfect motor-torque tracking (i.e., $\boldsymbol{\tau}_\mathrm{m} = \mathbf{\Omega}^\top \boldsymbol{\tau}_\mathrm{cmd}$), substituting (4) into (3) and projecting the dynamics back to the joint space gives

$$\boldsymbol{\tau}_\mathrm{net} = \boldsymbol{\tau}_\mathrm{cons} + \boldsymbol{\tau}_\mathrm{err}, \tag{6}$$

where $\boldsymbol{\tau}_\mathrm{cons}$ denotes the conservative reactions and $\boldsymbol{\tau}_\mathrm{err} = \boldsymbol{\tau}_\mathrm{fric} - \boldsymbol{\tau}_\mathrm{comp}$ represents the unmodeled friction, which is continuously corrected by the feedback controller ($\boldsymbol{\tau}_\mathrm{err} \approx \boldsymbol{\tau}_\mathrm{fb}$).

To isolate the non-conservative friction modeling error, we evaluate the cyclic energy balance of (6) over a closed periodic trajectory with period T:

$$\int_0^T \boldsymbol{\tau}_\mathrm{net}^\top \dot{\mathbf{q}}\,\mathrm{d}t = \int_0^T \boldsymbol{\tau}_\mathrm{cons}^\top \dot{\mathbf{q}}\,\mathrm{d}t + \int_0^T \boldsymbol{\tau}_\mathrm{err}^\top \dot{\mathbf{q}}\,\mathrm{d}t. \tag{7}$$

By the Work–Energy Theorem, the first integral vanishes over a closed cycle because the manipulator returns to its initial configuration with identical kinetic and potential energies:

$$\int_0^T \boldsymbol{\tau}_\mathrm{cons}^\top \dot{\mathbf{q}}\,\mathrm{d}t = 0.$$

To expose the directional effect of unmodeled cross-coupling, we evaluate the remaining non-conservative terms for a specific coordinate i. Equation (7) reduces to

$$E_{\mathrm{res},i} = \int_0^T \tau_{\mathrm{err},i}\,\dot{q}_i\,\mathrm{d}t = \int_0^T (\tau_{\mathrm{fric},i} - \tau_{\mathrm{comp},i})\,\dot{q}_i\,\mathrm{d}t. \tag{8}$$

This scalar $E_{\mathrm{res},i}$ quantifies the residual non-conservative work performed by the feedback controller for joint i over one motion period. Discrete-time controllers naturally inject artificial energy due to sampling delays; however, integrating over the macroscopic $2\,\mathrm{s}$ cycle isolates the controller effort to overcome persistent load-dependent friction $\boldsymbol{\tau}_\mathrm{err}$. We note that the metric relies on estimated velocities and Section IV describes the observer and quantifies the impact of estimation delays.

The sign of $E_{\mathrm{res},i}$ has a direct physical interpretation. If the decoupled friction model were accurate, $E_{\mathrm{res},i}$ would remain near zero regardless of the loading condition. Instead, positive values indicate that the transmission dissipates more energy than the controller compensates, increasing apparent friction and reducing transparency. Negative values mean the controller injects net energy over a closed trajectory. Because passive mechanical systems cannot generate energy, negative residual energy indicates that the compensator becomes an active element, reducing the available passivity margin [6].

TABLE I
IDENTIFIED STRIBECK FRICTION PARAMETERS.
UNITS: $F_{\mathrm{c},i}, F_{\mathrm{s},i}$ $[\mathrm{N}\cdot\mathrm{m}]$; $\dot{q}_{\mathrm{s},i}$ $[\mathrm{rad/s}]$; B_i $[\mathrm{N}\cdot\mathrm{m}/(\mathrm{rad/s})]$.

Joint i	$F_{\mathrm{c},i}$	$F_{\mathrm{s},i}$	$\dot{q}_{\mathrm{s},i}$	B_i
Joint 1	1.814	3.075	0.062	0.900
Joint 2	1.773	3.106	0.083	0.893

IV. EXPERIMENTAL METHODOLOGY

We now apply the energy-balance metric $E_{\mathrm{res},i}$ to the *SENTIR* differential haptic interface (Fig. 1), a custom three-degree-of-freedom device. To isolate the two-DoF differential stage, we removed the distal joint. The transmission uses a two-stage belt reduction and a straight bevel-gear differential with an overall ratio $R \approx 28.8$. We reconstruct joint positions from Hall-effect motor encoders using the kinematic transformation in (1).

The two joint coordinates excite distinct operating modes of the differential transmission. Differential motion (q_1) produces counter-rotating motor shafts, which rotate the carrier as a rigid body with negligible relative gear motion. Energy dissipation is dominated by the main support bearing. Common-mode motion (q_2) drives both motors in the same direction, inducing relative sliding between the bevel gears. Under this condition, tooth contact dominates energy dissipation. These modes provide distinct loading conditions under identical sensing and control architectures.

As baseline compensation, we implement a classical decoupled Stribeck friction model:

$$\tau_{\mathrm{comp},i} = \left[F_{\mathrm{c},i} + (F_{\mathrm{s},i} - F_{\mathrm{c},i})e^{-|\hat{v}_i|^{\eta}}\right]\mathrm{sgn}(\dot{q}_i) + B_i\dot{q}_i, \quad (9)$$

where $F_{\mathrm{c},i}$, $F_{\mathrm{s},i}$, and B_i are the Coulomb, static, and viscous friction parameters, respectively; $\eta = 2$ is the shape factor; $\hat{v}_i = \dot{q}_i/\dot{q}_{\mathrm{s},i}$ is the normalized velocity; and $\dot{q}_{\mathrm{s},i}$ is the Stribeck velocity.

Table I summarizes the model parameters identified from bidirectional constant-velocity sweeps covering the operating velocity range [16]. We identified symmetric (direction-independent) parameters to establish a conventional baseline, matching standard identification procedures [17]. An asymmetric decoupled model could slightly reduce single-axis residual errors but cannot capture friction variations at Joint i caused by load redistribution from Joint j.

Because only motor encoder measurements are available, we estimated joint velocities using an α-β-γ observer with parameters optimized through a multi-objective Pareto analysis balancing bandwidth, phase lag, noise sensitivity, and low-speed tracking performance (gains: $\alpha = 0.155$, $\beta = 0.00868$, $\gamma = 0.000162$). We evaluated the observer tracking performance against single-axis sinusoidal trajectories, yielding a Root Mean Square Error (RMSE) of $0.033\,\mathrm{rad/s}$ and $0.025\,\mathrm{rad/s}$ for Joints 1 and 2, respectively, and a $2\,\mathrm{ms}$ maximum cross-correlation delay. We held the identified friction parameters and observer gains fixed throughout all experiments, ensuring that residual energy variations reflect exclusively changes in cross-coupled loading conditions.

TABLE II
PERIODIC EXCITATION TRAJECTORIES. ALL TRAJECTORIES SHARE $A = 30^\circ$ AND $\omega = \pi\,\mathrm{rad/s}$.

Experiment	Joint 1	Joint 2
1	$A\sin(\omega t)$	0
2	0	$A\sin(\omega t)$
3	$A\cos(\omega t)$	$A\sin(\omega t)$
4	$A\sin(\omega t)$	$A\sin(\omega t)$
5	$A\sin(\omega t)$	$-A\sin(\omega t)$

We evaluated five periodic trajectories (Table II) to progressively increase cross-axis excitation. All experiments included a mandatory $1\,\mathrm{min}$ warm-up at nominal operating speeds to minimize thermal drift. Cycle-to-cycle stationarity was verified across consecutive runs, indicating negligible friction drift due to thermal variations over the reported cycles.

Experiments 1 and 2 excite pure differential and common modes, establishing baselines. Experiment 3 uses a Lissajous trajectory for simultaneous excitation. Experiments 4 and 5 use in-phase and counter-phase diagonal motions, which kinematically isolate one motor while the other drives the entire transmission. All experiments used amplitude $A = 30^\circ$, frequency $f = 0.5\,\mathrm{Hz}$ ($\omega = \pi\,\mathrm{rad/s}$), and a controller sampling period of $1\,\mathrm{ms}$. To avoid high-frequency inertial forces and structural resonances that disturb friction estimation, we selected this quasi-static $0.5\,\mathrm{Hz}$ excitation regime that isolates the measured energy deviations to load-dependent sliding friction.

At this frequency, the observer's maximum cross-correlation delay of $\Delta t = 2\,\mathrm{ms}$ corresponds to a phase shift of $\delta\phi = 2\pi f \Delta t \approx 0.0063\,\mathrm{rad}$ (0.1% of the $2\,\mathrm{s}$ excitation period). To assess the sensitivity of the residual energy to this delay, we consider the resulting integration error from (8):

$$\Delta E_{\mathrm{res},i} = \int_0^T \tau_{\mathrm{err},i}(t)\left(\dot{q}_i(t) - \dot{\hat{q}}_i(t)\right)\mathrm{d}t. \quad (10)$$

Assuming a dominant harmonic velocity profile $\dot{q}_i(t) = V\cos(\omega t)$, the delayed estimate is $\dot{\hat{q}}_i(t) = V\cos(\omega t - \delta\phi)$. Applying the trigonometric addition formula and the small-angle approximations $\cos(\delta\phi) \approx 1$ and $\sin(\delta\phi) \approx \delta\phi$ yields $\dot{\hat{q}}_i(t) \approx V\cos(\omega t) + V\delta\phi\sin(\omega t)$, so the velocity estimation error is $\dot{q}_i(t) - \dot{\hat{q}}_i(t) \approx -V\delta\phi\sin(\omega t)$. The resulting observer-induced energy error is

$$\Delta E_{\mathrm{res},i} \approx -V\delta\phi\int_0^T \tau_{\mathrm{err},i}(t)\sin(\omega t)\,\mathrm{d}t. \quad (11)$$

The error amplitude is proportional to $\delta\phi$ relative to the nominal velocity amplitude V. Based on this relationship, the relative error in the evaluated energy integral is assumed to satisfy $|\Delta E_{\mathrm{res},i}|/|E_{\mathrm{res},i}| \le \delta\phi$. Thus, a phase shift of $0.0063\,\mathrm{rad}$ corresponds to an estimated observer-induced energy calculation error of approximately 0.63% of the true $E_{\mathrm{res},i}$.

The elastic belt transmission acts as a conservative energy-storage mechanism, also making its additional phase lag

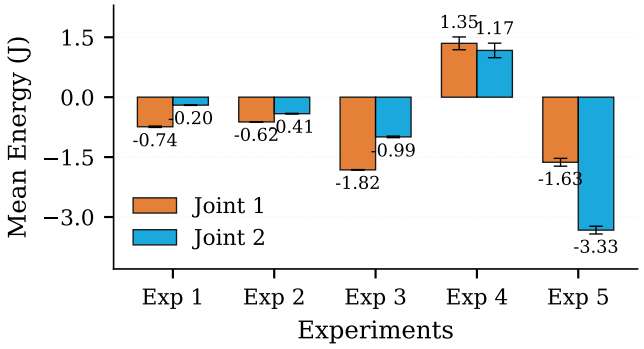


Fig. 2. Residual non-conservative energy $E_{\mathrm{res},i}$ per cycle for each experiment.

negligible under these quasi-static conditions. Zero-mean estimation noise and localized non-conservative transients (e.g., backlash impacts) are expected to produce bounded fluctuations that largely cancel over a full cycle. These estimates suggest that any significant variation in $E_{\mathrm{res},i}$ can be attributed to structural load-dependent friction coupling rather than transmission compliance or sensing effects.

V. RESULTS

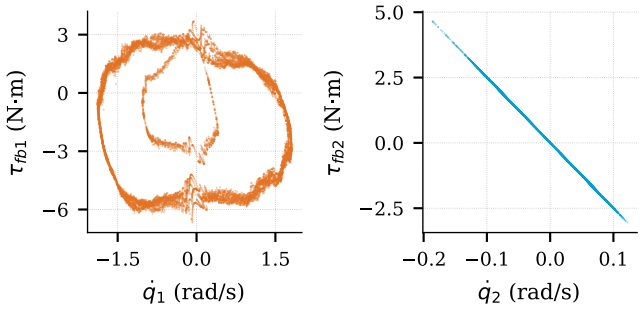
To account for system variability, all reported $E_{\mathrm{res},i}$ values represent the mean and standard deviation computed over five consecutive steady-state cycles.

A. Energy-Balance Analysis

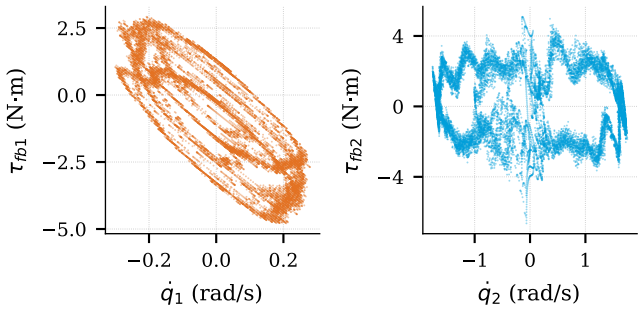
Evaluating the residual non-conservative energy $E_{\mathrm{res},i}$ per cycle indicates that the identified Stribeck model overestimates friction even under single-axis conditions (Fig. 2). In Experiment 1, $E_{\mathrm{res},1} = -0.74 \pm 0.02\,\mathrm{J}$ and $E_{\mathrm{res},2} = -0.20 \pm 0.01\,\mathrm{J}$. In Experiment 2, the corresponding values were $-0.62 \pm 0.005\,\mathrm{J}$ and $-0.41 \pm 0.008\,\mathrm{J}$. These negative residual energy values confirm that the model injects a small amount of active energy. Because the compensator models are fixed, any subsequent variation in $E_{\mathrm{res},i}$ reflects the energetic impact of the cross-coupling.

The mismatch grows under simultaneous motion. During the Lissajous trajectory (Experiment 3), $E_{\mathrm{res},i}$ drops to $-1.82 \pm 0.007\,\mathrm{J}$ and $-0.99 \pm 0.02\,\mathrm{J}$, representing 2.46 and 2.41 times the pure-mode values, respectively. This decrease implies that the transmission dissipates less energy than the isolated-joint model predicts, causing increased over-compensation.

The largest deviations occur during the diagonal trajectories. In Experiment 5, $E_{\mathrm{res},2} = -3.33 \pm 0.1\,\mathrm{J}$, nearly eight times the magnitude observed in the pure common-mode excitation. This amplification indicates severe over-compensation, injecting over 3 J of active energy per cycle. At this magnitude, the energy injection far exceeds any millijoule-level uncertainty introduced by the observer phase lag or unmodeled conservative compliance, potentially reducing the system's passivity margin.



(a) Pure differential mode (Experiment 1).



(b) Pure common mode (Experiment 2).

Fig. 3. Residual torque hysteresis during (a) pure differential and (b) common-mode excitations. The stationary joints exhibit cross-coupled residual torque despite negligible motion.

Experiment 4 displays the opposite behavior, with both joints exhibiting positive residual energies of $1.35 \pm 0.16\,\mathrm{J}$ and $1.17 \pm 0.18\,\mathrm{J}$ per cycle. Positive $E_{\mathrm{res},i}$ implies that the decoupled model underestimates friction under this specific loading condition, acting as a dissipative bottleneck that degrades haptic transparency.

The large spread in $E_{\mathrm{res},i}$, from active injection ($-3.33\,\mathrm{J}$) to uncompensated dissipation ($+1.35\,\mathrm{J}$), suggests that baseline tuning offsets cannot explain the mismatch. No single set of diagonal parameters could stabilize this system across all quadrants, confirming the inability of decoupled models to capture load-dependent friction.

B. Residual Torque Analysis

These energy deviations originate from structured residual torque hysteresis that appears even during single-joint evaluations (Fig. 3). The excited joints exhibit hysteresis loops consistent with a reasonably compensated friction model, which explains why joint-wise identification is often considered satisfactory. The stationary joints, in contrast, develop deterministic residual torques reaching $5\,\mathrm{N}\cdot\mathrm{m}$ despite negligible motion.

These residual torques cannot be attributed to local friction modeling errors, as the corresponding joints remain nearly stationary. Instead, they arise from the internal torque redistribution required to constrain one degree of freedom while actuating the other. Their amplitude is highly sensitive to the initial preload adjustment (higher preloads increase the baseline normal forces redistributed by the cross-coupled

torque), but the loop shapes are insensitive to mechanical backlash. The preload prevents severe zero-crossing transients, preserving geometrically continuous hysteresis without the flattened deadbands typical of high backlash. Because the feedforward Stribeck term produces negligible torque for the stationary joint, the feedback controller must reject these cross-coupled loads omitted by the decoupled formulation. Although these residual torques produce only moderate energy deviations during pure motions, they explain the imbalance observed during multi-axis trajectories.

C. Directional Anisotropy of Residual Torque

To isolate the directional dependence of transmission friction from velocity-magnitude effects, we exploit the geometric properties of the Lissajous trajectory (Experiment 3). This excitation produces a joint-velocity vector $\dot{\mathbf{q}}(t)$ that traces a circular path in joint-velocity space with constant norm:

$$\|\dot{\mathbf{q}}(t)\| = \sqrt{\dot{q}_1^2(t) + \dot{q}_2^2(t)} = A\omega \approx 1.64\,\mathrm{rad/s}. \tag{12}$$

Because the velocity magnitude is fixed, variations in $\boldsymbol{\tau}_{\mathrm{net}}$ arise from the direction of motion and the associated redistribution of internal loads. The instantaneous torque includes conservative dynamic components, but their effects remain small at the quasi-static frequency of $0.5\,\mathrm{Hz}$, such that the residual torque predominantly reflects friction compensation errors. We define the instantaneous velocity direction $\theta_{\mathrm{vel}} \in [-\pi, \pi]$ as

$$\theta_{\mathrm{vel}}(t) = \mathrm{atan2}\left(\dot{q}_2(t), \dot{q}_1(t)\right), \tag{13}$$

and the residual torque magnitude $\|\boldsymbol{\tau}_{\mathrm{net}}\|$ as

$$\|\boldsymbol{\tau}_{\mathrm{net}}(t)\| = \sqrt{\tau_{\mathrm{net},1}^2(t) + \tau_{\mathrm{net},2}^2(t)}. \tag{14}$$

Expressing these quantities in polar coordinates reveals the directional structure of the transmission residual torque (Fig. 4). Under the decoupled assumption, the distribution of $\|\boldsymbol{\tau}_{\mathrm{net}}\|$ would be radially symmetric, indicating direction-independent compensation error. Instead, the residual torque ranges from a baseline of $0.05\,\mathrm{N\cdot m}$ to a peak of $8.6\,\mathrm{N\cdot m}$, exhibiting an asymmetric multi-lobed pattern directly linked to (1).

The residual torque is smallest near the diagonal regions ($\theta_{\mathrm{vel}} \approx 45°, 135°, 225°, 315°$), where the joint velocities satisfy $\dot{q}_1 = \pm\dot{q}_2$. Under these conditions, one motor remains stationary. With only one actuator and its corresponding reduction stage in motion, internal mechanical shear and gear-meshing losses are minimized, making the transmission behavior consistent with the decoupled friction model.

The largest residual torque occurs along the principal axes ($\theta_{\mathrm{vel}} \approx 0°, 90°, 180°, 270°$), where both motors rotate simultaneously. When only the second joint moves ($\theta_{\mathrm{vel}} \approx 90°$, $\dot{q}_1 = 0$), both motors rotate at the same speed and in the same direction, increasing internal mechanical shear and axial loading. The peak residual torque reaches approximately $8.6\,\mathrm{N\cdot m}$ in this direction, compared with $4.2\,\mathrm{N\cdot m}$ for the opposite motion ($\theta_{\mathrm{vel}} \approx 270°$).

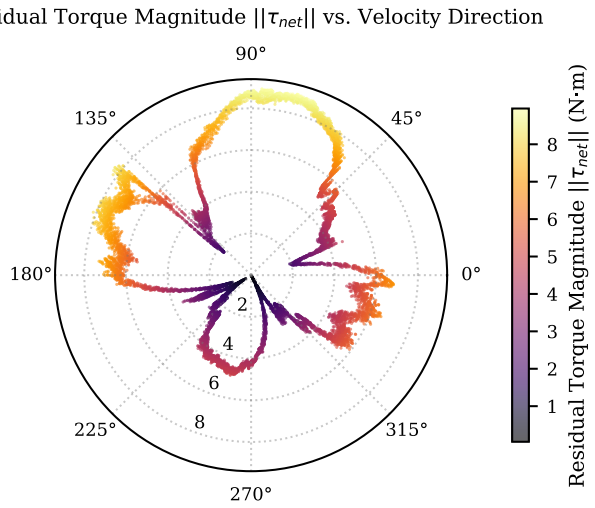


Fig. 4. Polar mapping of the residual torque magnitude $\|\boldsymbol{\tau}_{\mathrm{net}}\|$ as a function of the joint-velocity direction angle θ_{vel} during Experiment 3 (constant velocity norm $\|\dot{\mathbf{q}}\| \approx 1.64\,\mathrm{rad/s}$).

This asymmetry suggests that the residual friction depends on the internal loading and power-flow direction. In gear transmissions, Coulomb friction depends on the normal force at the active contact interface, which varies with the transmitted torque and the engaged interface. Reversing the power flow changes the active interface and redistributes internal axial and thrust loads, altering the normal force even at a constant velocity magnitude. Kikuuwe [18] modeled this behavior using a differential-algebraic inclusion with input-side and output-side asymmetry coefficients to capture load-dependent friction differences between the input and output shafts. Dohring et al. [19] derived a similar effect for a wedge-like worm-gear model, showing that the active contact interface determines the steady-state force relation and yields distinct friction expressions for forward and reverse power flow. Their experiments confirmed that power-flow direction strongly affects frictional losses. Wang and Kim [20] similarly characterized directional efficiency, demonstrating that a gear pair operates at different efficiencies in step-down and step-up modes.

Our joint-space compensator, as used in the literature, does not account for these load and power-flow dependencies, leaving the associated friction uncompensated and producing the asymmetric residual torque lobes in Fig. 4. These findings indicate that, under the tested conditions, decoupled friction models produce significant errors when attempting to capture load-induced structural asymmetries.

VI. CONCLUSION

This work shows that decoupled friction compensation can induce active, asymmetric behavior in differential transmissions. The cyclic energy-balance analysis reveals that fixed-gain diagonal models fail to capture internal load redistribution during multi-axis motion, resulting in both substantial energy

injection ($-3.33\,\mathrm{J}$ per cycle) and uncompensated dissipation ($+1.35\,\mathrm{J}$).

These results characterize the controller's energy behavior under continuous sliding friction at $0.5\,\mathrm{Hz}$ but do not establish its stability during human interaction. The human arm exhibits a frequency-dependent mass-spring-damper impedance whose damping can absorb injected energy and potentially prevent sustained vibrations [21], [22], [23]. The "biomechanical excess of passivity" quantified in [24] highlights the need to analyze the coupled human-robot dynamics to determine whether the arm can dissipate the injected energy at relevant interaction frequencies ($\approx 10\,\mathrm{Hz}$) [25], [26]. Further experiments are needed to assess the effects of frequency transients and validate the findings with other transmission architectures, such as cable-driven differentials.

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